

Program : Diploma in Wood and Paper Technology	
Course Code : 2121	Course Title: Botany and Anatomical Structures
Semester : 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:3, T:0, P:0)	Periods per semester: 45

Course Objectives:

- To impart knowledge on the different types of classifications in the plant kingdom
- To familiarize with the plants having immense economic importance.
- To impart an insight into the internal structure and reproduction of plants

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in Botany		Secondary School	

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize plant diversity	8	Understanding
CO2	Describe the plant taxonomy and morphology of plant parts	13	Understanding
CO3	Illustrate plant anatomy and plant tissues	12	Understanding
CO4	Discuss the Plant body structure and wood anatomy	10	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize plant diversity		
M1.01	Outline Plant Classification	1	Remembering
M1.02	List the salient features of algae and fungi	2	Remembering
M1.03	Explain the classification , structure & economic importance of the main classes of bryophytes, pteridophytes and gymnosperms	3	Understanding
M1.04	Discuss the inheritance pattern of nuclear and extra nuclear genes	2	Understanding
Contents: Diversity of life: two kingdom classification (Carolus Linnaeus, 1735); phylogenetic classification (August W Eichler, 1878);criteria for classification, general characters of each kingdom Diversity of plants: study the salient features of algae, fungi, bryophytes, pteridophytes, gymnosperms and its economic importance Ultrastructure of nucleus and it's importance - genetic material - DNA - genes - gene action - change in gene structure leading to plant diversity			
CO2	Describe the plant taxonomy and morphology of plant parts		
M2.01	Illustrate the leaf and flower morphology	4	Understanding
M2.02	Explain inflorescence	3	Understanding
M2.03	Give the significance of plant taxonomy	3	Understanding
M2.04	Explain the vegetative and floral characteristics of economically important plants within the families	3	Understanding
	Series Test - I	1	

Contents:**Pant morphology**

Parts of a typical flowering plant (Eg., Root system - tap root, fibrous root, adventitious root) Leaves - Parts of a leaf - simple leaf - compound leaf - phyllotaxy,

Inflorescence - definition - classification - recemose - cymose - special type - parts of a flower, type of flowers -floral whorls- placentation - pollination - fertilization.

Angiosperm taxonomy:-significance, Binomial nomenclature - description and identifying characters of the following families with emphasize on economically important plants.

1. Annanoceae - Annona, Polyalthia
2. Fabeaceae - Tamarindus, Accasia
3. Rubiaceae - Ixora, Hamelia
4. Verbanacea - Latana, Tectona
5. Poaceae - Paddy, Wheat
6. Malvaceae - Gossypium, Hibiscus
7. Euphorbiacea - Rubber
8. Myrtaceae - Euclalyptussps.

Herbarium technique

CO3	Illustrate plant anatomy and plant tissues		
M3.01	Explain the Structure and composition of plant cells	6	Understanding
M3.02	Give the characteristic features, functions and classification of tissues	6	Understanding
Contents: Cell - Ultra structure, living and non-living inclusions; cell wall - Ultra structure - pits - plasmodesmata. Tissues - Meristematic and Permanent meristems - Definition - General characters - Types of meristems, based on position - origin shoot apex - root apex - theories of meristems - permanent tissues - classification - simple tissues - complex tissues. Vascular bundles - origin - types - conjoint - collateral - open and closed – eccentric - concentric - amphivasal - amphicribal - radial.			
CO4	Discuss the Plant body structure and wood anatomy		
M4.01	Explain the primary structure of stem, root and leaves	4	Understanding
M4.02	Explain the secondary structure of stem and root	3	Understanding
M4.03	Illustrate wood anatomy	3	Understanding
	Series Test - II	1	

Contents:

Primary structure of Dicot stem (Eupatorium), Monocot stem (Grass), Dicot root (Tinospora), Monocot root (Colocasia) Structure of dicot leaf (Ixora), Structure of monocot leaf (Grass) Normal secondary growth in dicot stem - cambial ring - cambial activity secondary xylem - secondary phloem growth rings - soft wood -hard wood - sap wood-heart wood - tyloses - structure - periderm formation - phellem - phellogen - phelloderm - lenticels functions

Text / Reference:

T/R	Book Title/Author
T1	B.R.Vashishta, Botany for Degree students Fungi,S Chand publication, 2014.
R1	B.R.Vashishta Botany for Degree students, Bryophyte:, Chand publication, 2014.
R2	B.R.Vashishta, Botany for Degree students, Algae:, Chand publication, 2014.
R3	B.P.Pandey Plant anatomy:, Chand Publishing, 2012
R4	B.P.Pandey, Economic Botany:.Chand Publication.1999
R5	Singh&Jain, Taxonomy: Rastogi Publications
R6	College Botany Vol.I, II, III: New central books agency, 2011

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=K2teJ6-DBLw&ab_channel=CEC20BT11
2	https://nptel.ac.in/content/storage2/courses/122103039/pdf/mod1.pdf
3	https://nptel.ac.in/courses/102/107/102107075/
4	https://nptel.ac.in/courses/122/103/122103039/